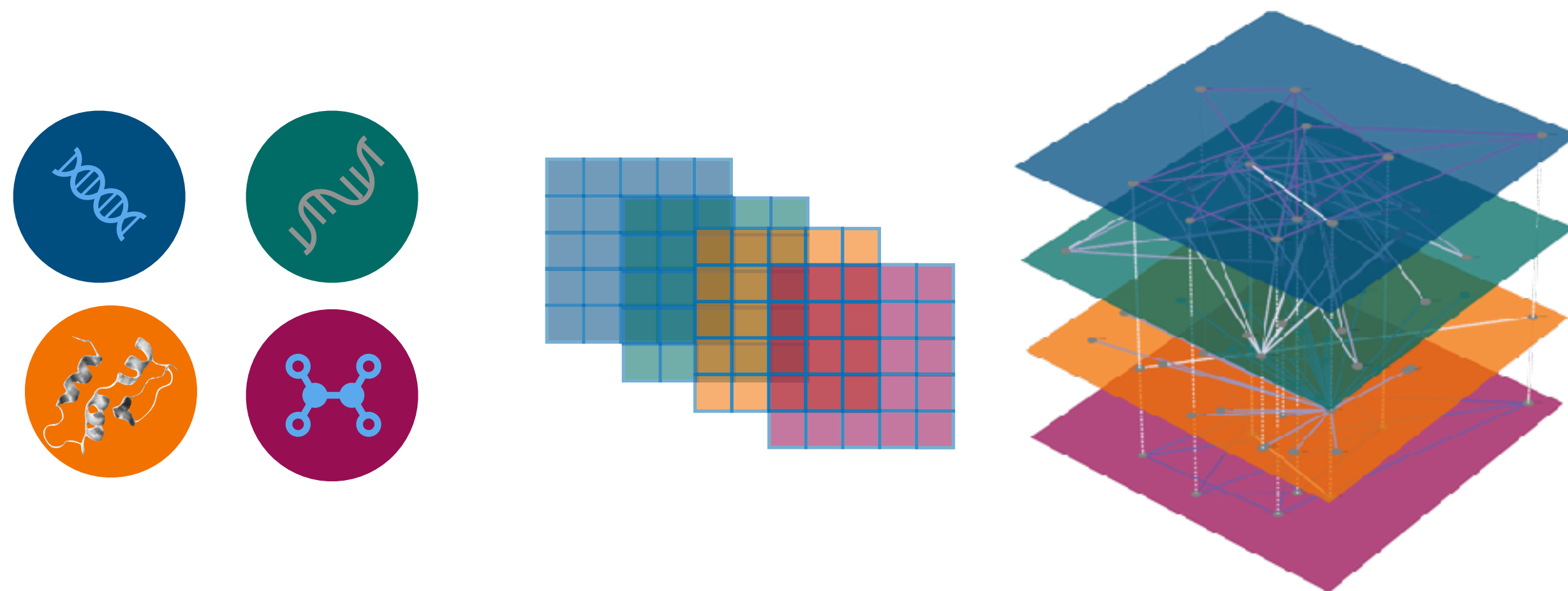




Graph Machine Learning

AI-guided Protein Science

Alberto Santos — Multi-omics Network Analytics (MoNA)

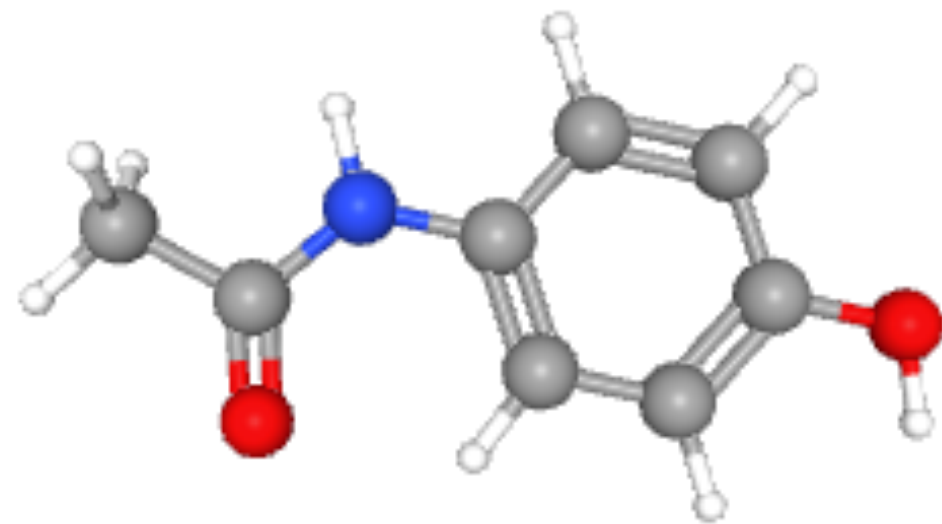
DTU Health Tech
Department of Health Technology



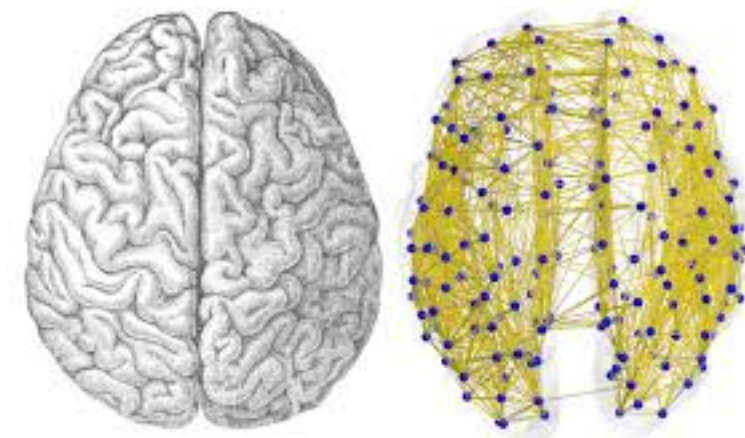
Limitations of Traditional Deep Learning

- Standard DL (CNNs, RNNs, Transformers) is limited to work with data that is **structured** in regular formats, such as **images**, **sequences**, and **grids**.
- What about **different structures**?

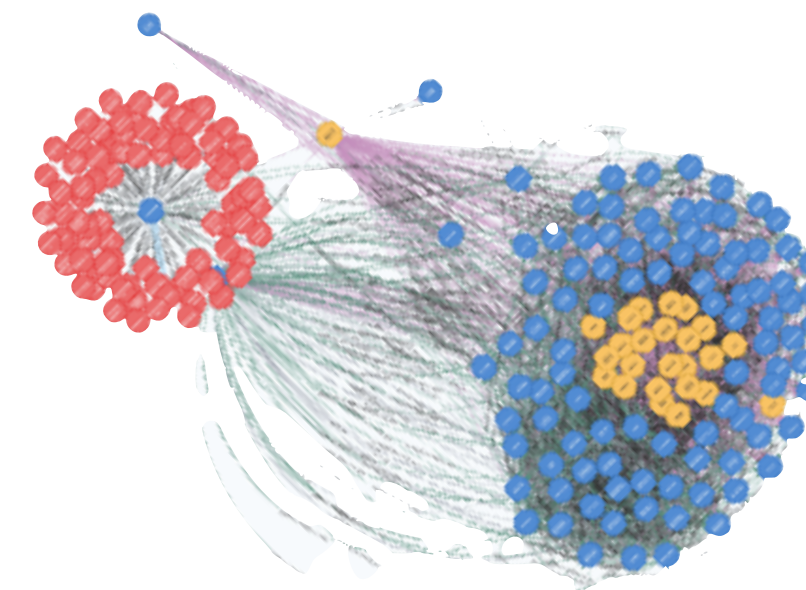
Graphs are everywhere and help describe complex systems



Molecules



Connectivity



Knowledge graphs



PPIs

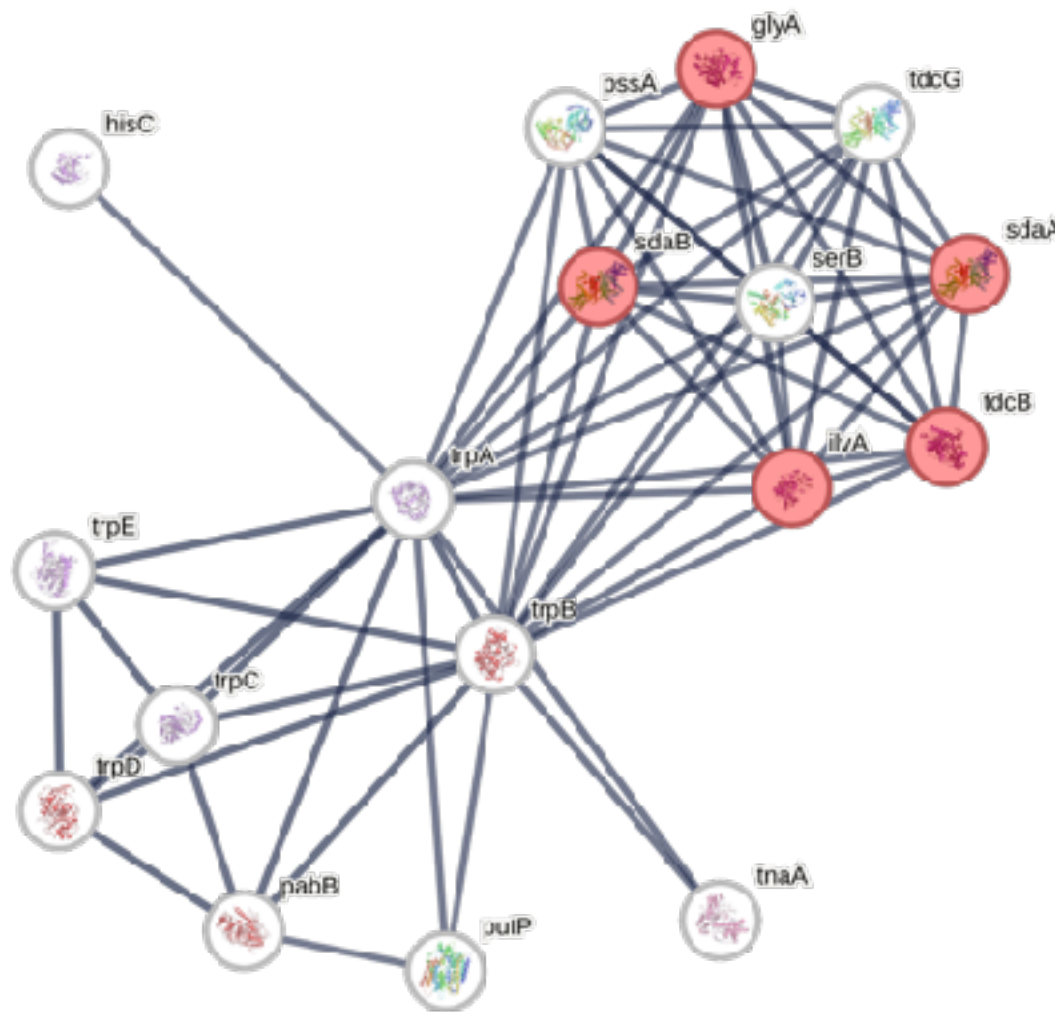
Machine Learning on Graphs

- Three types of prediction tasks:
 - **Node-level**: predict the identity, role or features of nodes within a graph.
 - **Edge-level**: predict relationships between nodes or features of these relationships.
 - **Graph-level**: predict the property of an entire graph.
- Graphs contain different information that we can use to make predictions: nodes, edges, global-context and connectivity.

Node Classification

- Predict the label (type, category or attribute) associated with all the nodes in the graph

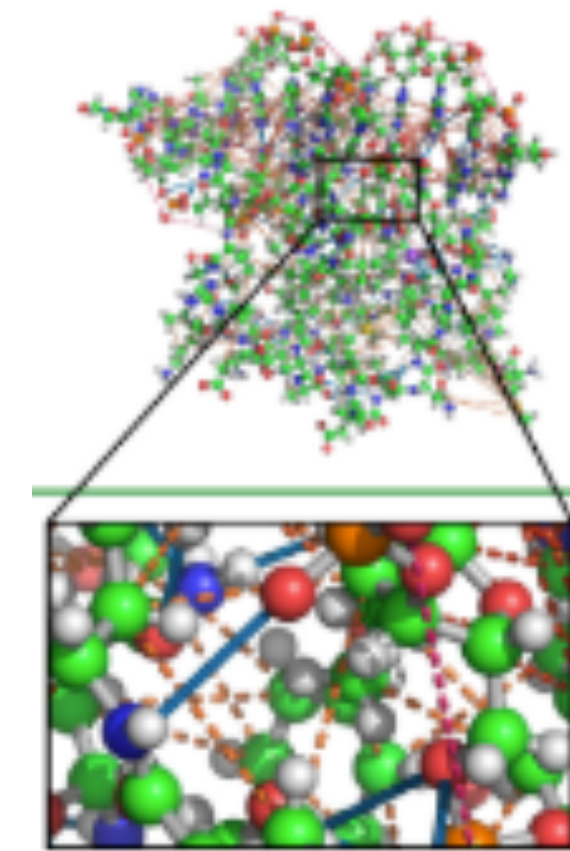
Examples



Classifying the function of proteins



Classifying cell types

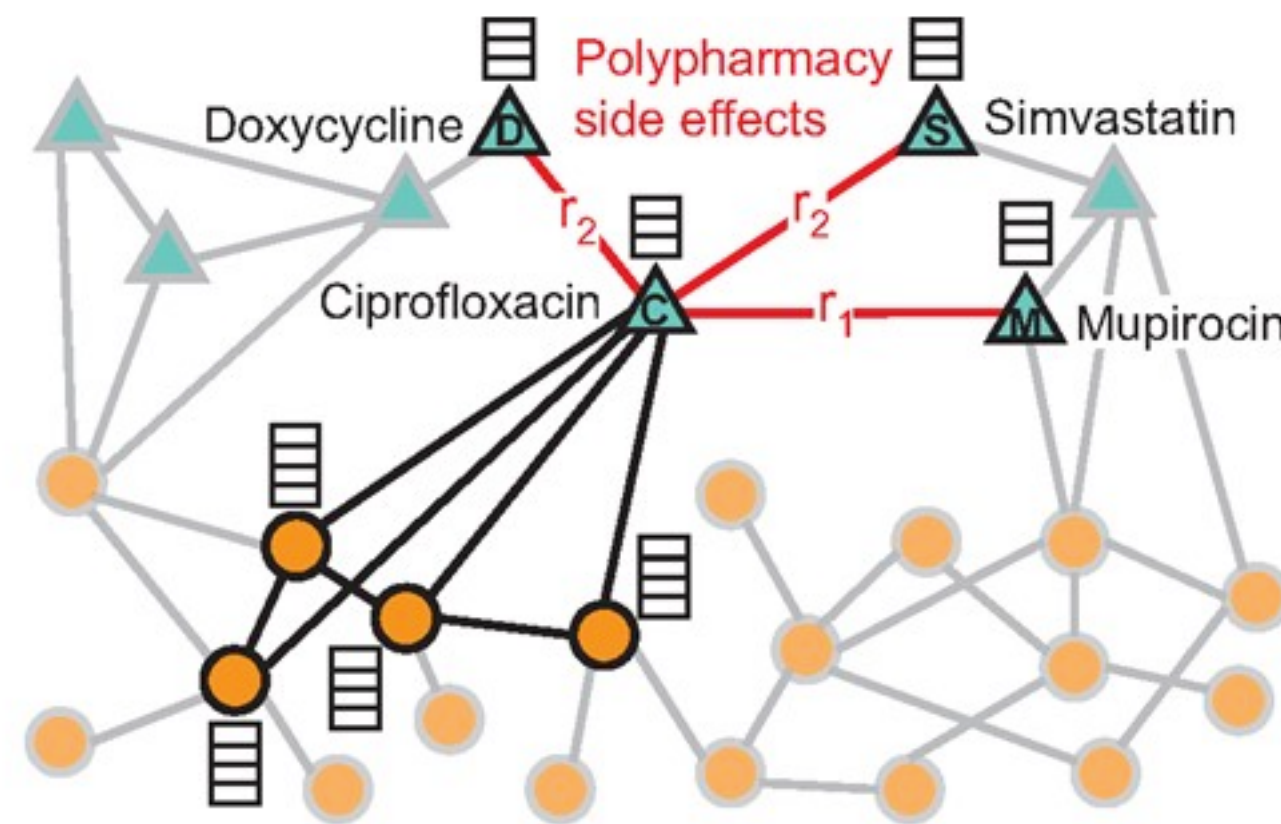


Classifying amino acids

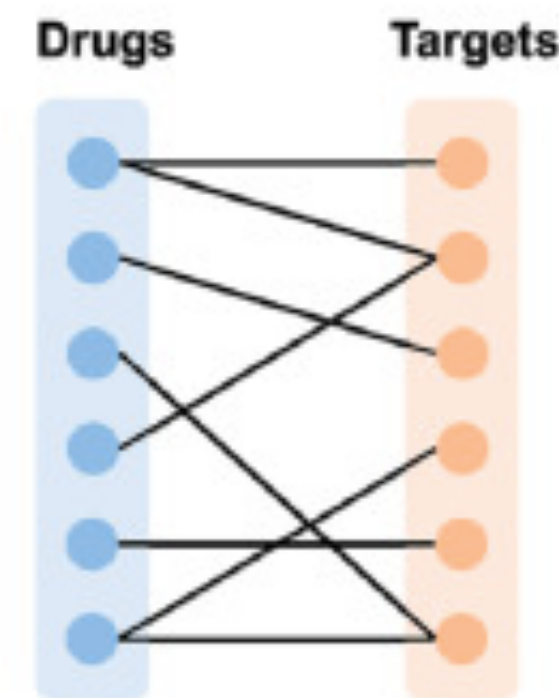
Relation Prediction

- Given a set of nodes and an incomplete set of edges between these nodes, we want to infer the missing edges using the structure of the graph

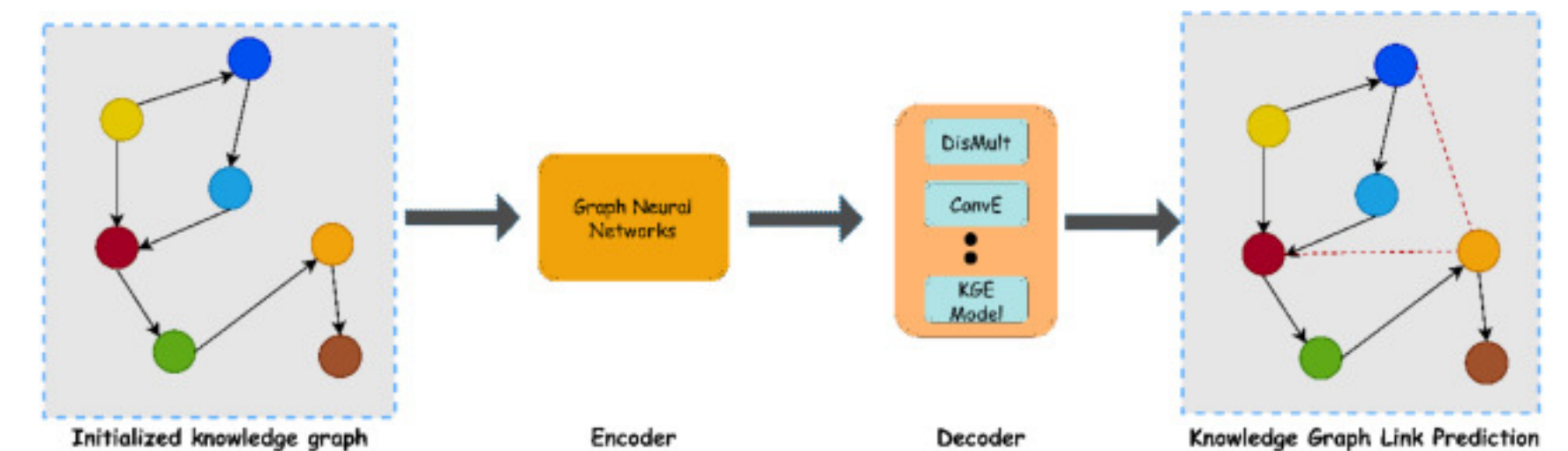
Examples



Drug-Side effect prediction



Drug-Target prediction



Knowledge Graph completion

Modeling polypharmacy side effects with graph convolutional networks. 2018. Marinka Zitnik, Monica Agrawal, Jure Leskovec

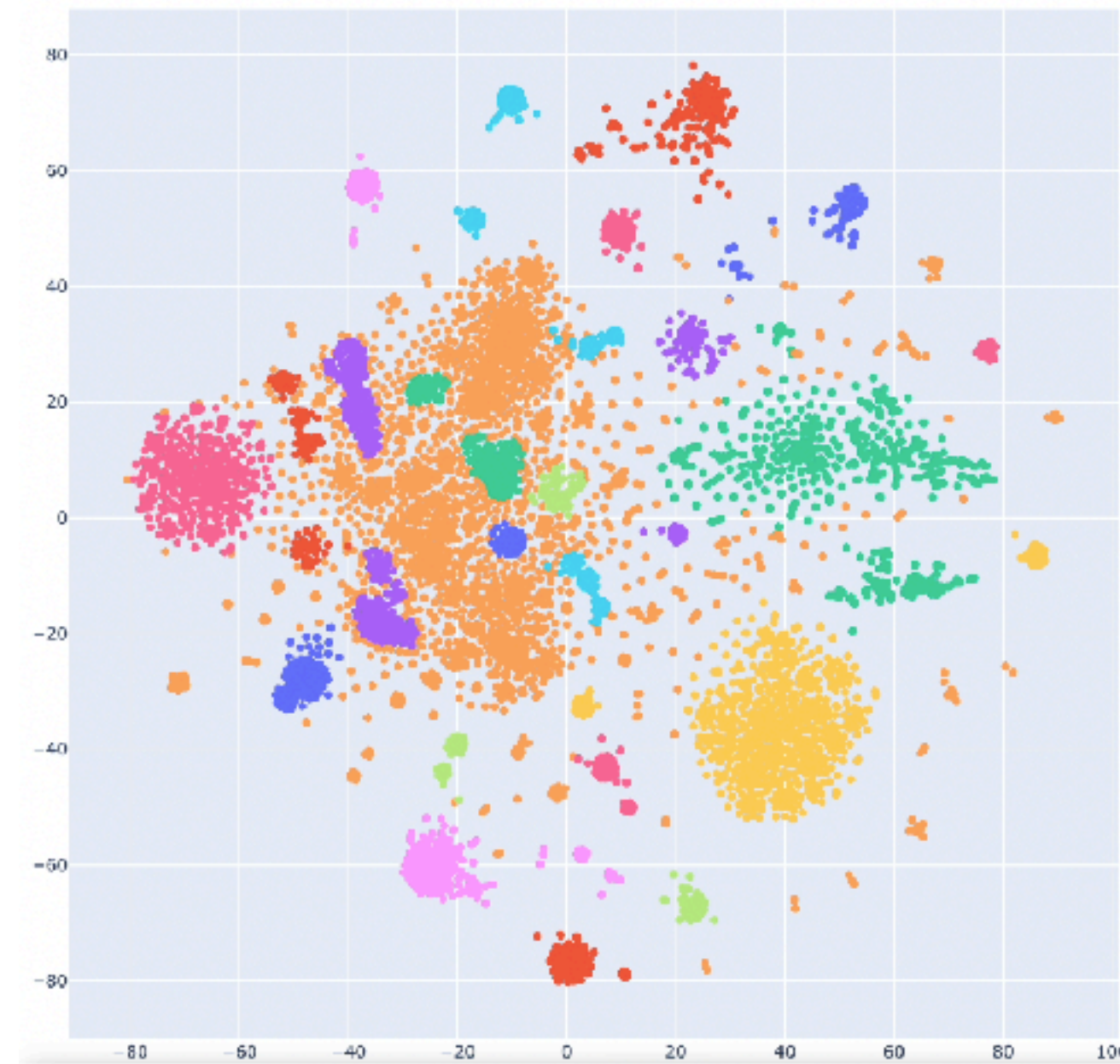
Graph neural network approaches for drug-target interactions. 2022. Zehong Zhang, Lifan Chen, Feisheng Zhong, Dingyan Wang, Jiaxin Jiang, Sulin Zhang, Hualiang Jiang, Mingyue Zheng, Xutong Li

A knowledge graph completion model based on contrastive learning and relation enhancement method.2022. LinYu Li, Xuan Zhang, YuBin Ma, Chen Gao, Jishu Wang, Yong Yu, Zihao Yuan, Qiuying Ma

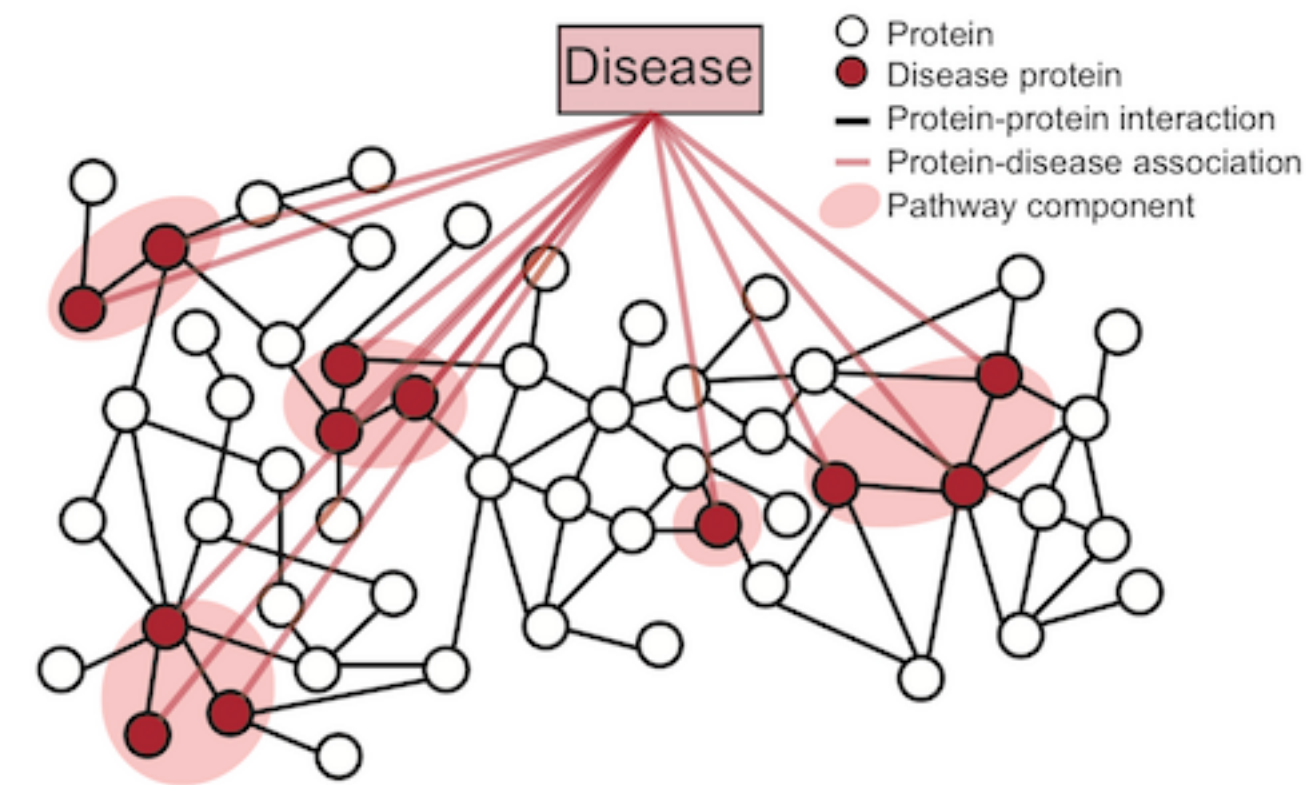
Community Detection

- Identify clusters where nodes are more likely to form edges

Examples



Detecting microbial communities



Disease pathways

Large-scale analysis of disease pathways in the human interactome. 2018. Monica Agrawal, Marinka Zitnik, Jure Leskovec.

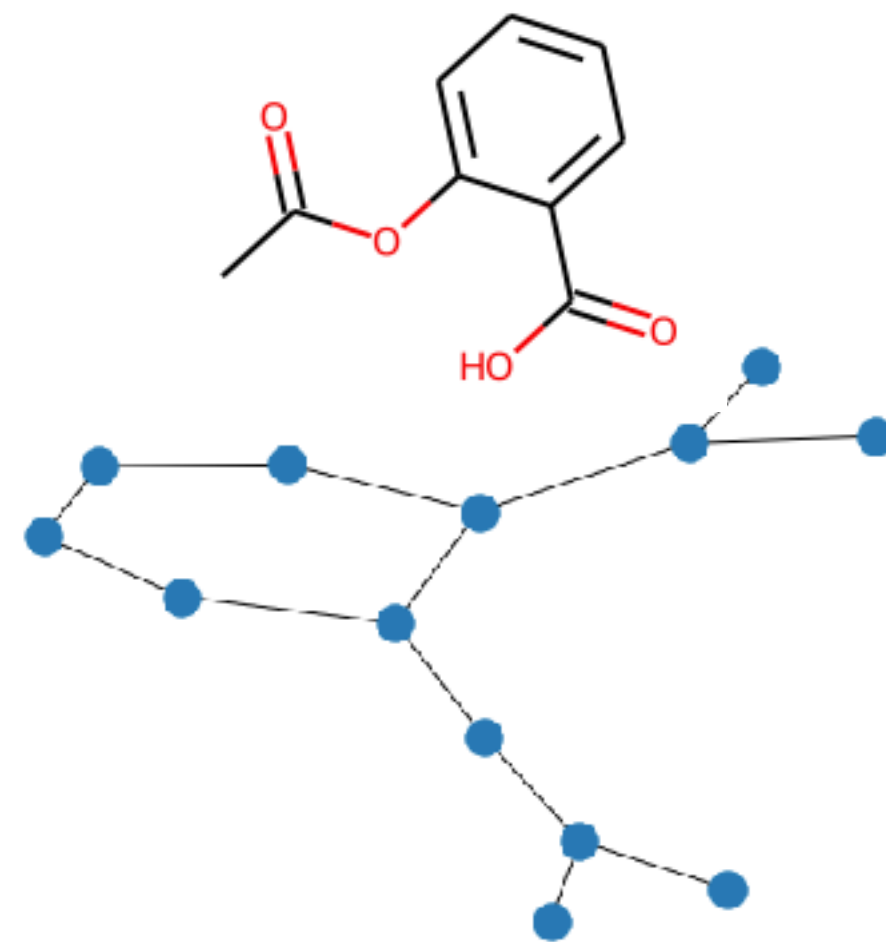
https://micw2graph.streamlit.app/Microbial_association_networks

https://github.com/benedekrozemberczki/awesome-community-detection/blob/master/chapters/deep_learning.md

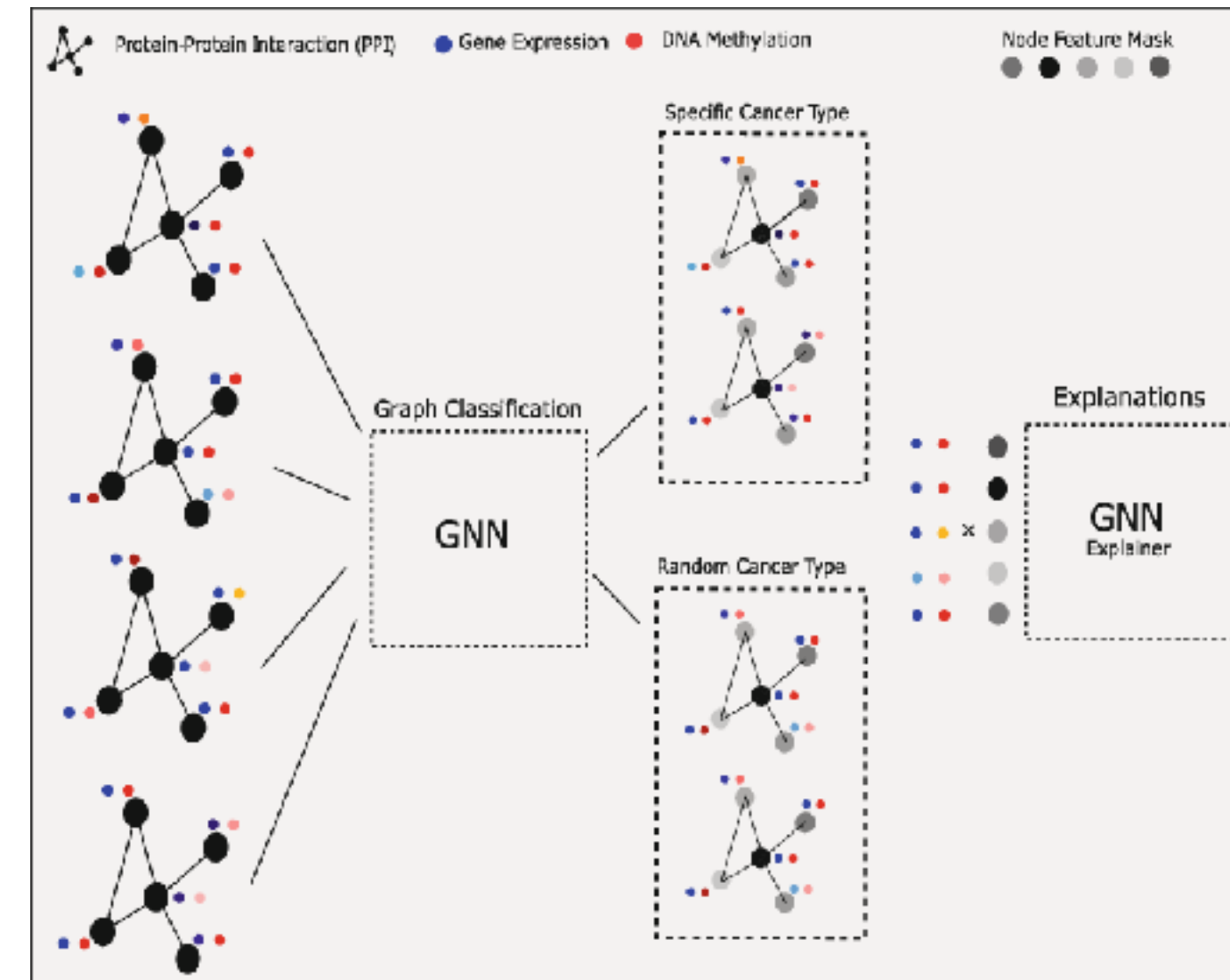
Graph Classification/Regression

- Learn over graph data and make independent predictions for each graph

Examples



Predicting molecular properties



Patient stratification

Key Concepts in GML

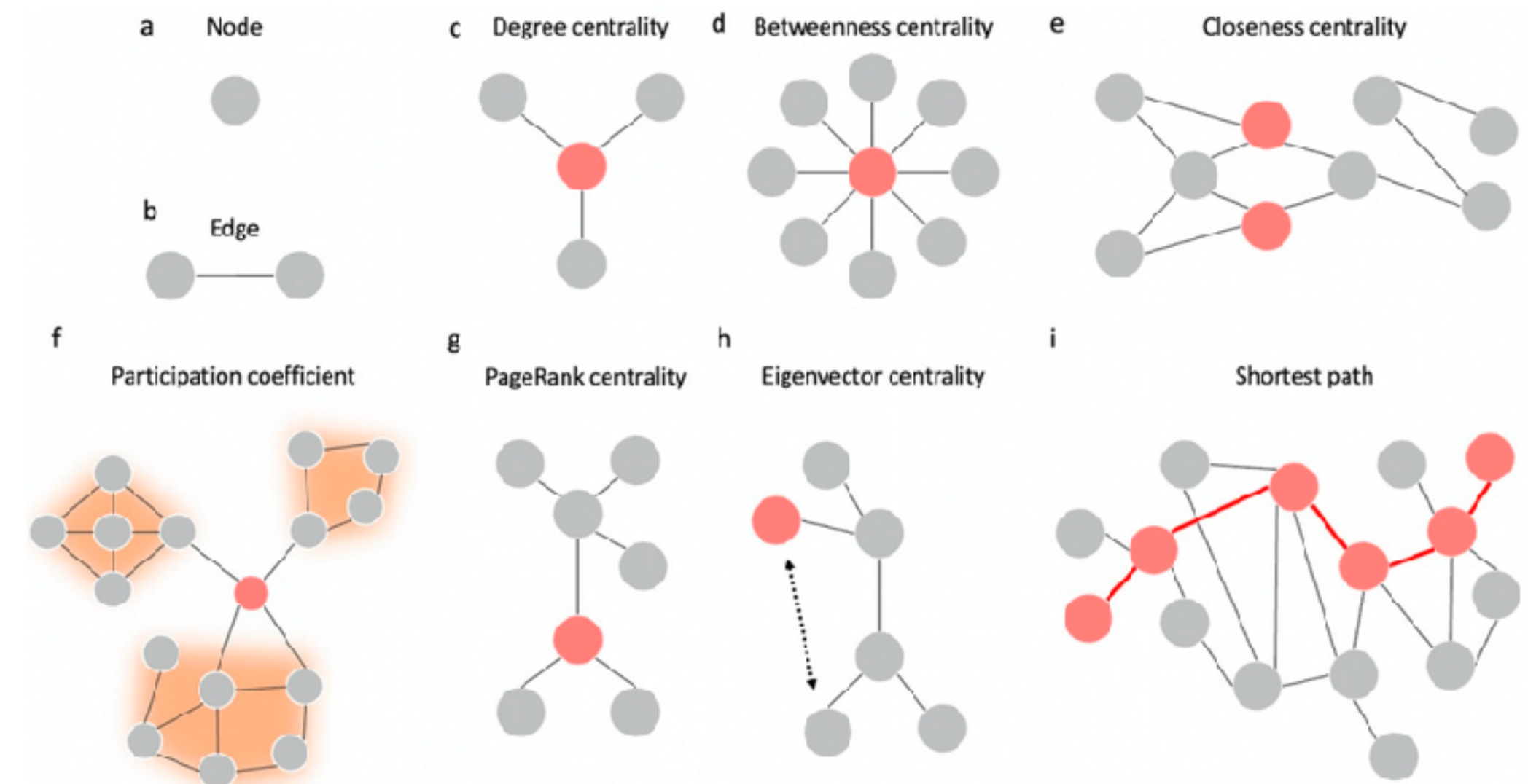
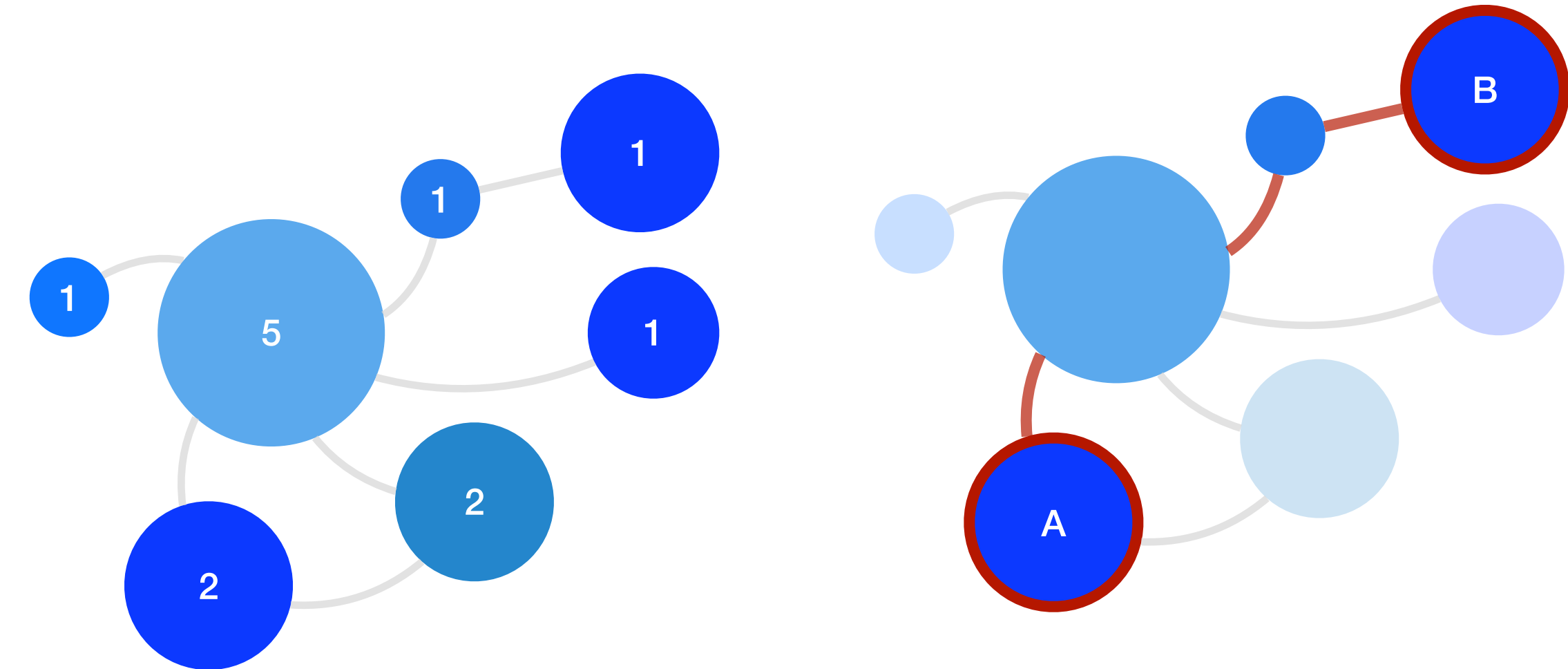
- **Interdependence** — points are not independent and identically distributed (interconnected nodes)
- **Homophily** — tendency of nodes to share attributes with their neighbours
- **Structural equivalence** — Nodes with similar local structures will share similar labels

Traditional GML Approaches

Extracting some **statistics or hand-crafted features** from the graph and using these features as input to an standard machine learning classifier

Node-level statistics

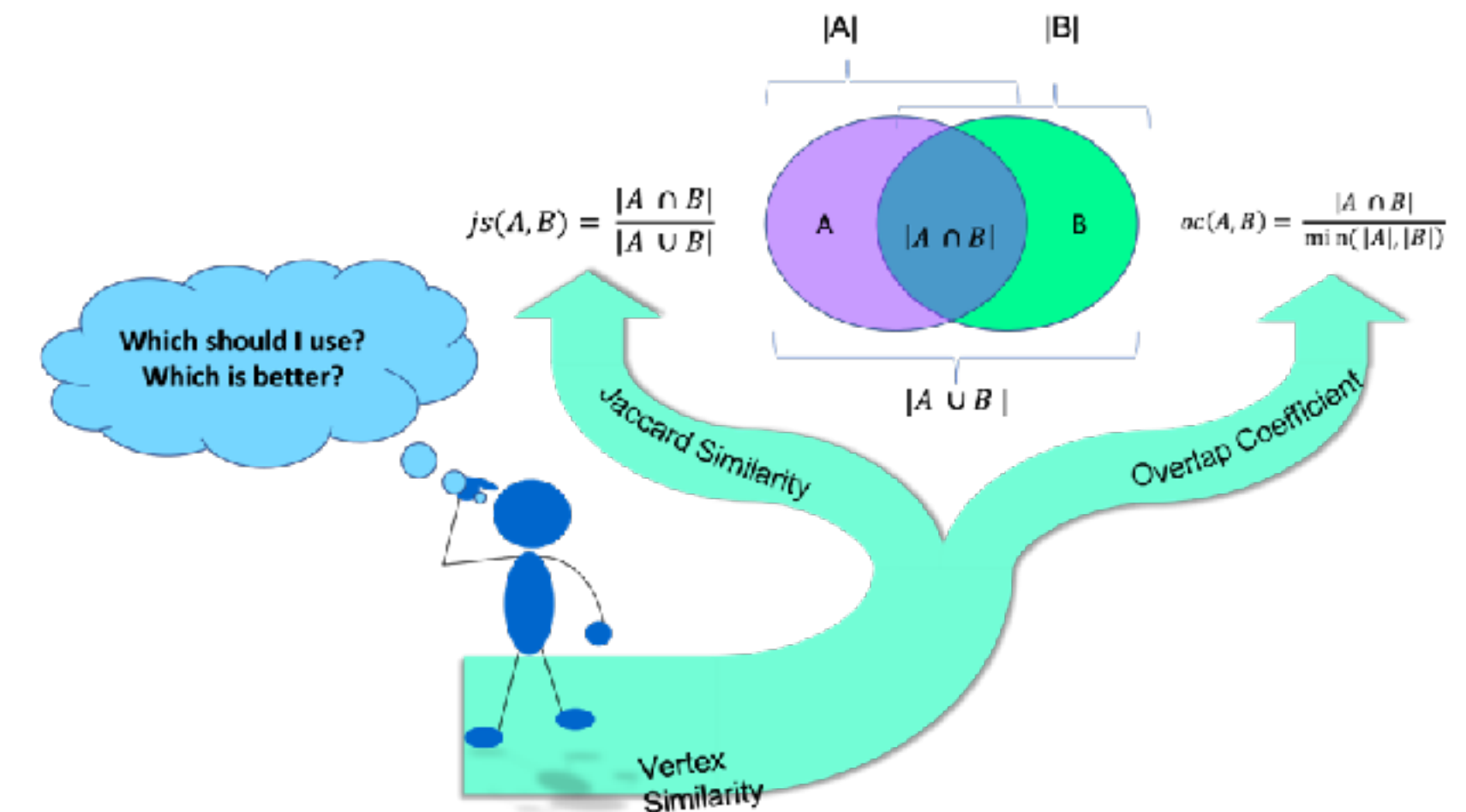
- Node degree
- Node centrality
- Clustering coefficient



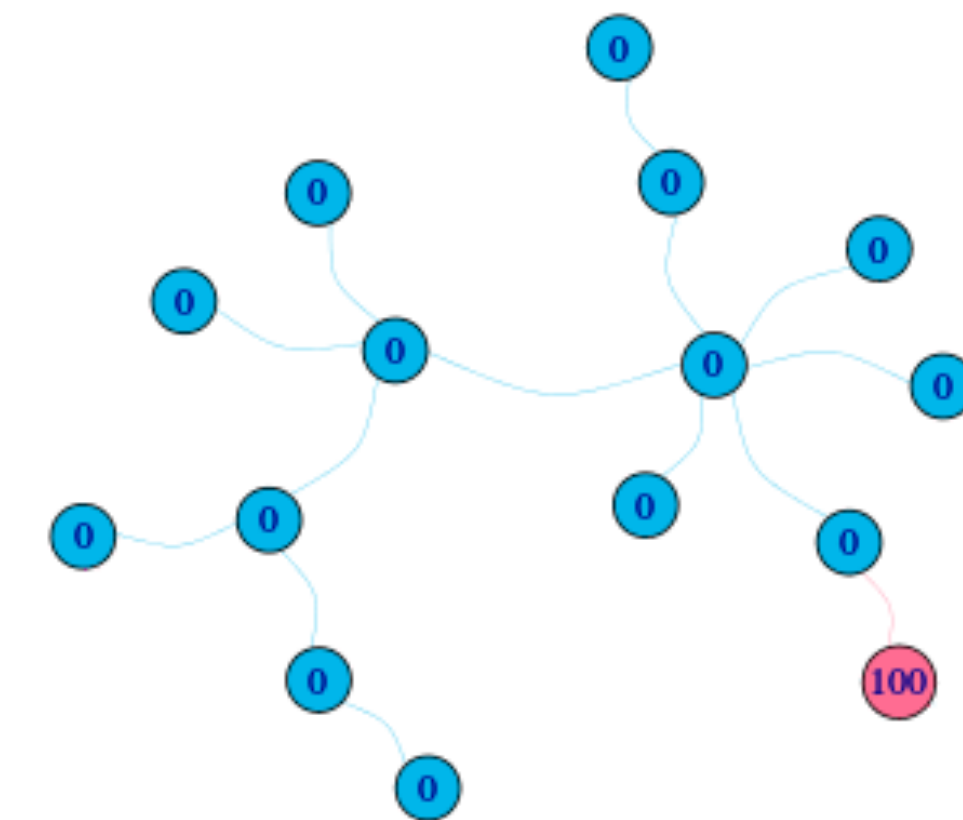
Traditional GML Approaches

Edge-level statistics

- Local overlap: Number of common neighbours two nodes share (e.g., Jaccard overlap)
- Global overlap:
 - *Katz Index*: number of paths of all lengths between two nodes. The higher the total score, the closer two nodes are globally.
 - *Leicht, Holme, and Newman Similarity*: ratio between the number of observed paths and the number of paths between two nodes expected by random chance given the size of the network.
 - *Random walk methods*: considering random walks instead of exact counts of paths (e.g., Personalised PageRank, probability that a random walk starting at one node visits another).



Walk length: 0 Alpha: 0.1 Distance: Inf



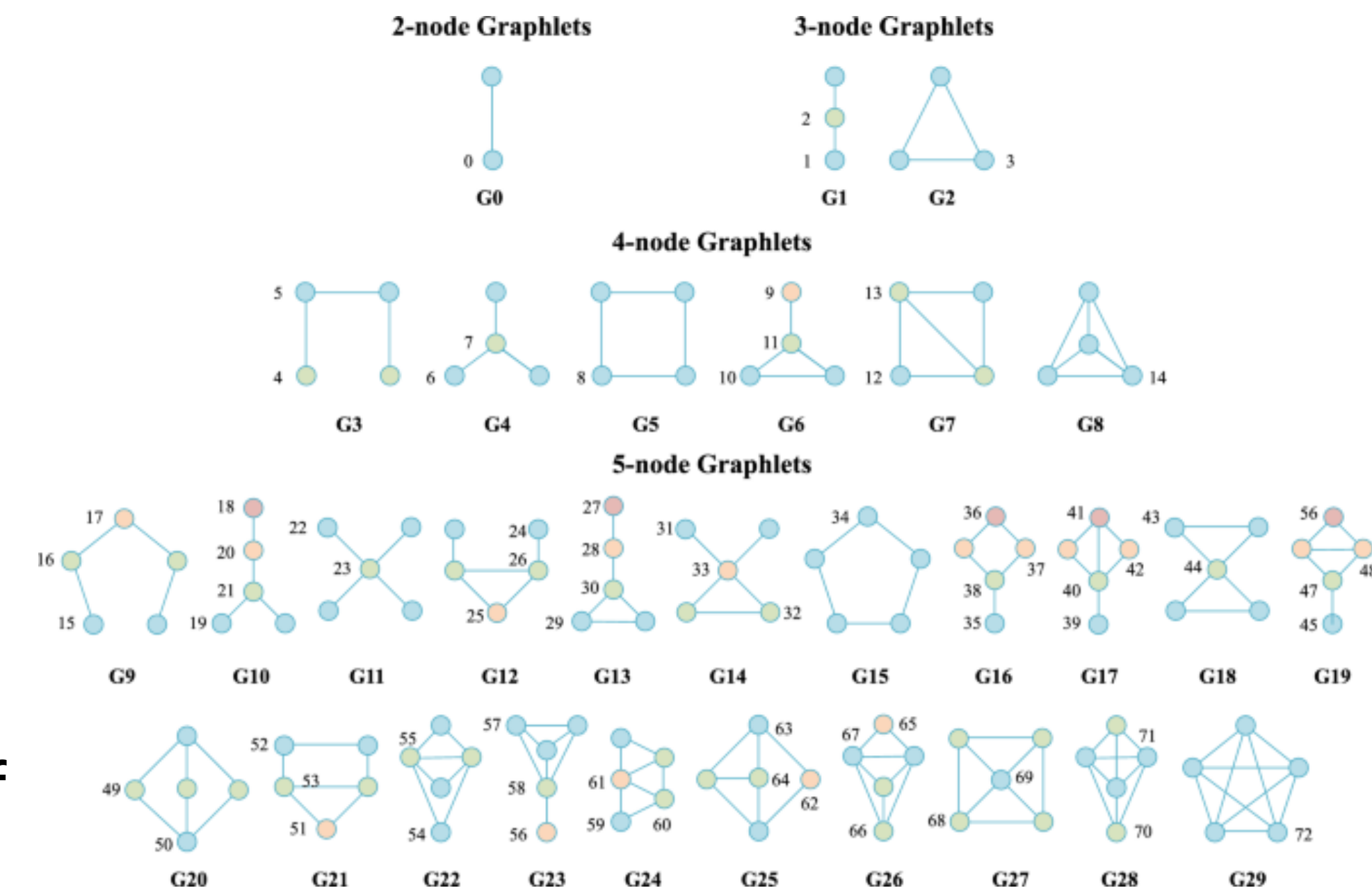
Personalised PageRank

<https://www.r-bloggers.com/2014/04/from-random-walks-to-personalized-pagerank/>

Traditional Approaches

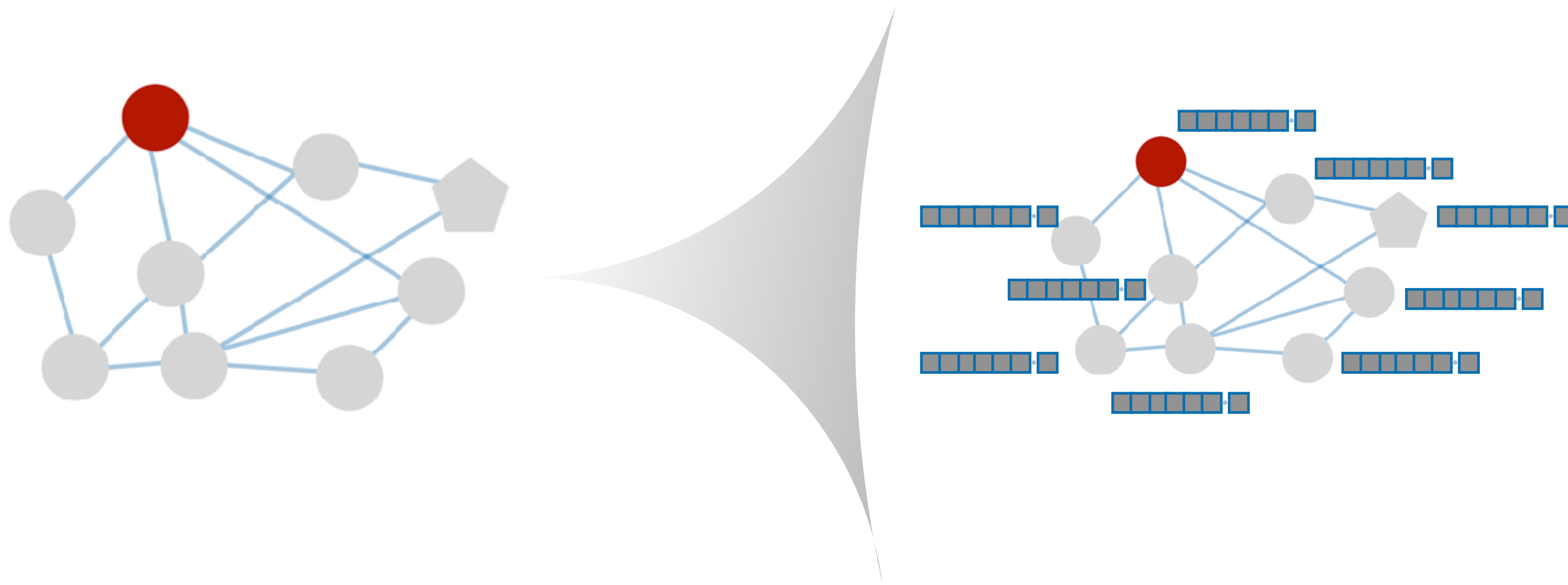
Graph-level statistics

- **Bag of nodes:** aggregating node-level statistics (summary statistics based on node stats)
- **Iterative Neighbour Aggregation:** extract node-level features that contain rich information about their neighbours and aggregate it at the graph-level — e.g., The Weisfieler-Lehman Kernel
- **Graphlets:** count the occurrence of different small subgraph structures (graphlets)
- **Path-based methods:** evaluates the different kinds of paths that occur in the graph (e.g., random walks/ shortest paths counting occurrence of sequences of node labels (degree))



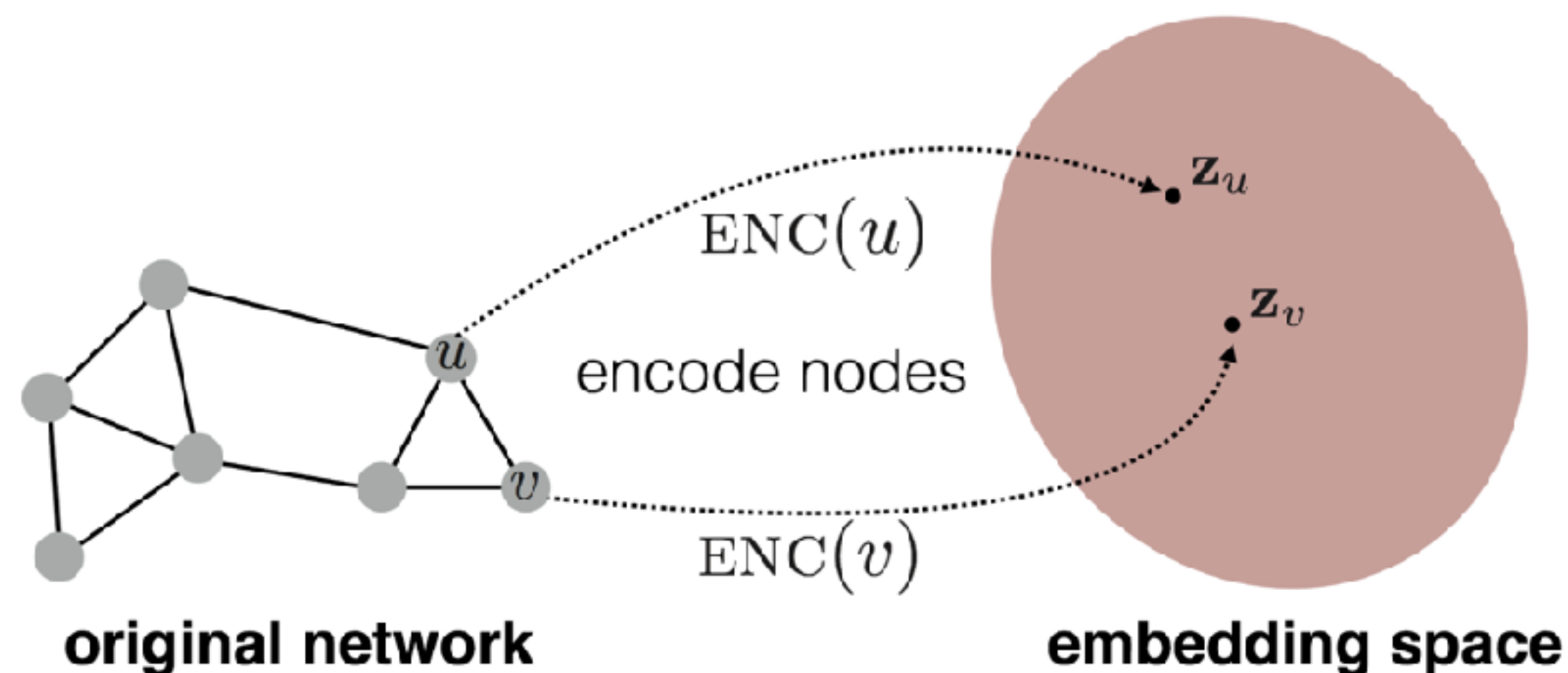
Graph Representation Learning

- The traditional approaches are limited
- Graph Representation Learning — learn representations that encode structural information about the graph to use them downstream



Node Embeddings

- Encode nodes as low-dimensional vectors summarising their position and structure of the local graph neighbourhood
- Similarity in the embedding space (e.g., dot product) approximates similarity in the original graph
- Encoder-decoder methods:



Encoder-Decoder Methods

Encoder: goal learn a mapping from nodes to embeddings. We Define a node similarity function and optimise the parameters of the encoder so that similarity

$$S[u, v] \approx \mathbf{z}_u^T \mathbf{z}_v$$

Decoder: reconstruct certain graph statistics from the node embedding (e.g., pairwise decoders predict whether two nodes are neighbours in the graph).

$$\text{DEC}(\mathbf{z}_u, \mathbf{z}_v) \approx S[u, v]$$

Shallow Embedding Methods

Type	Method	Decoder	Similarity measure	Loss function
Matrix Factorisation	Laplacian Eigenmaps	$\ z_u - z_v\ _2^2$	General	$\text{DEC}(z_u, z_v) \cdot S[u, v]$
	Graph Factorisation	$z_u^\top z_v$	$A[u, v]$	$\ \text{DEC}(z_u, z_v) \cdot S[u, v]\ _2^2$
	GraRep	$z_u^\top z_v$	$A[u, v], A^2[u, v], \dots, A^k[u, v]$	$\ \text{DEC}(z_u, z_v) \cdot S[u, v]\ _2^2$
	Hope	$z_u^\top z_v$	General	$\ \text{DEC}(z_u, z_v) \cdot S[u, v]\ _2^2$
Random walk	DeepWalk	$\frac{e^{z_u^\top z_v}}{\sum_{k \in \mathcal{V}} e^{z_u^\top z_k}}$	$p_{\mathcal{G}}(v \mid u)$	$-S[u, v] \cdot \log(\text{DEC}(z_u, z_v))$
	node2vec	$\frac{e^{z_u^\top z_v}}{\sum_{k \in \mathcal{V}} e^{z_u^\top z_k}}$	$p_{\mathcal{G}}(v \mid u)$ (biased)	$-S[u, v] \cdot \log(\text{DEC}(z_u, z_v))$

$p_{\mathcal{G}}(v \mid u)$ Probability of visiting v on a fixed length random walk starting from u

Random Walk Methods

- Two nodes have similar embeddings if they cooccur on short random walks

Type	Method	Decoder	Similarity measure	Loss function
Random walk	DeepWalk	$\frac{e^{z_u^\top z_v}}{\sum_{k \in \mathcal{V}} e^{z_u^\top z_k}}$	$p_{\mathcal{G}}(v \mid u)$	$-S[u, v] \cdot \log(\text{DEC}(z_u, z_v))$
	node2vec	$\frac{e^{z_u^\top z_v}}{\sum_{k \in \mathcal{V}} e^{z_u^\top z_k}}$	$p_{\mathcal{G}}(v \mid u)$ (biased)	$-S[u, v] \cdot \log(\text{DEC}(z_u, z_v))$

- These methods differ on how they overcome the complexity of evaluating the loss function ($O(|V|)$).

Limitations of Shallow Embeddings

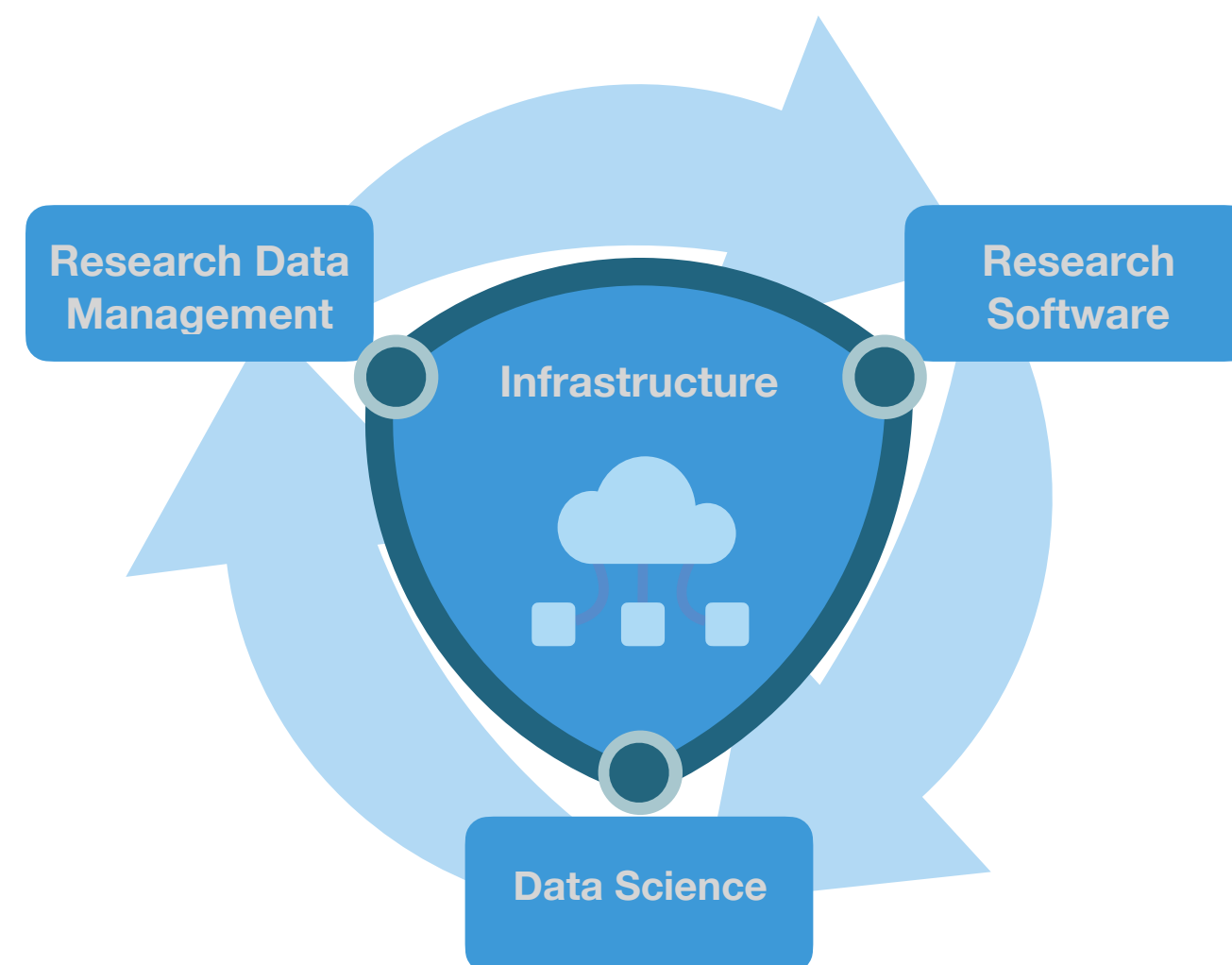
- **Complexity grows** with the number of nodes ($O(|V|)$) — problem with large graphs
- They **do not use node features**
- They are **transductive** — can only generate embeddings for nodes present during the training phase (no generalizable to unseen nodes)

Thank you

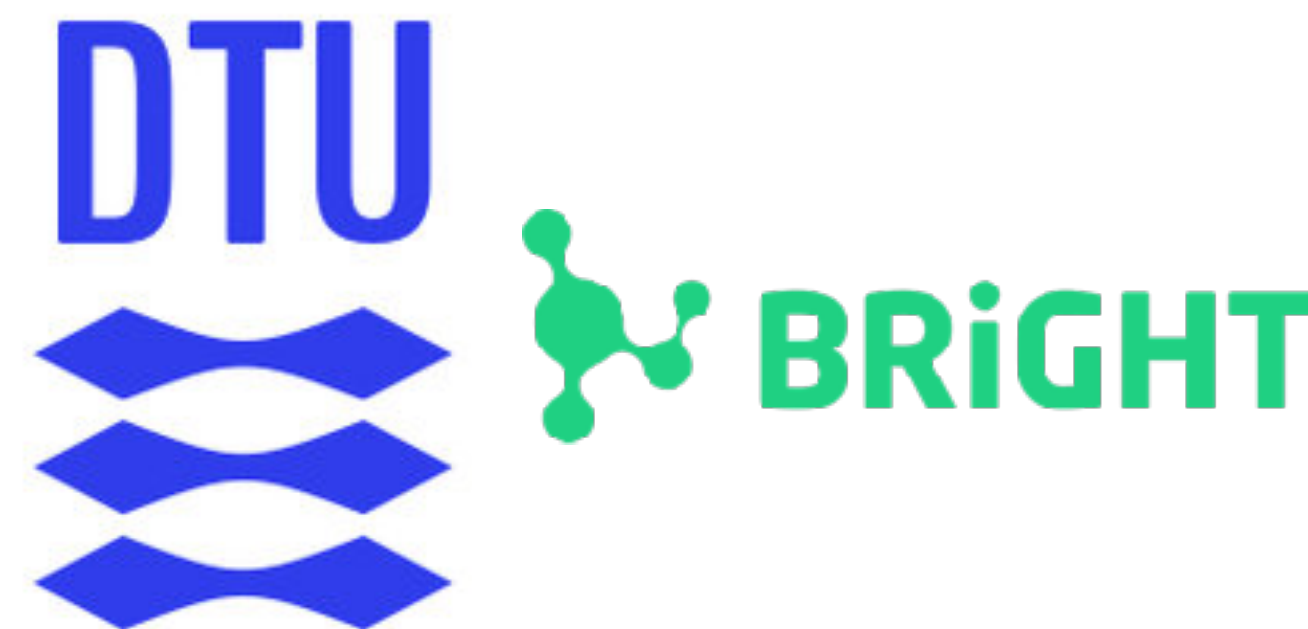
Multi-omics Network Analytics Research Group



Informatics Platform



DTU Health Tech
Department of Health Technology



albsad@dtu.dk



@albsantosdel



<https://github.com/Multiomics-Analytics-Group>



<https://multiomics-analytics-group.github.io/>



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